

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

A customer-level clustering study using Orange Data Mining

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PROGRAM	Computer Science & Artificial Intelligence (CSAI)
YEAR	3rd Year
SECTION	B
ROLL NO.	87
INSTITUTION	JECRC Foundation
ACADEMIC YEAR	2026–27

Academic Year 2026–27

PROJECT AT A GLANCE

Dataset UCI Online Retail	Transactions 541,909	Customers 4,372
Method K-Means	Clusters 3	Evaluation Silhouette Analysis

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1. Introduction

Customer segmentation is the process of dividing customers into groups with similar characteristics or purchasing behaviour. In this project, K-Means clustering is used to segment customers from the UCI Online Retail dataset. The project is implemented using Orange Data Mining.

2. Problem Statement

The objective is to analyse online retail transaction data and group customers into meaningful segments based on their purchasing activity. Such segmentation can help identify regular, high-activity and higher-value customer groups.

3. Objectives

- Load and inspect the Online Retail dataset.
- Handle missing values using preprocessing.
- Aggregate transaction-level data at customer level.
- Select relevant numerical features for clustering.
- Standardize the features so that different scales do not dominate clustering.
- Apply K-Means clustering with 3 clusters.
- Evaluate the clusters using a Silhouette Plot and visualize them using a Scatter Plot.

4. Dataset Description

The project uses the UCI Online Retail dataset. It contains 541,909 transaction records from a UK-based online retail business covering transactions from December 2010 to December 2011. Important fields include InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country.

5. Data Preprocessing

Missing values were handled using Orange's Impute Missing Values preprocessing step. The transaction-level data was then grouped by CustomerID to create one record per customer. The aggregation used Sum for Quantity, Mean for UnitPrice, Count for InvoiceNo and Count for StockCode. InvoiceDate and Country were not used as clustering features.

6. Feature Selection and Standardization

The final clustering features were Quantity-Sum, UnitPrice-Mean, InvoiceNo-Count and StockCode-Count. CustomerID was retained as a meta attribute. The selected numerical features were standardized using $\mu = 0$ and $\sigma^2 = 1$ so that features with larger numerical ranges would not dominate the K-Means distance calculation.

7. K-Means Clustering

K-Means clustering was applied with a fixed number of 3 clusters. KMeans++ initialization was used, with 10 re-runs and a maximum of 300 iterations. Normalization of columns was enabled and the model was applied automatically.

8. Cluster Results

The customer-level dataset contained 4,372 customers after grouping. The three clusters showed different behavioural patterns. Cluster 1 had a strong silhouette score and represented the more central/regular customer group. Cluster 2 showed high activity and volume characteristics, with high standardized Quantity-Sum and InvoiceNo/StockCode counts. Cluster 3 was strongly distinguished by a much higher

standardized UnitPrice-Mean, indicating a higher-price/premium-price group.

9. Silhouette Analysis

The Silhouette Plot was used to evaluate how well customers fit within their assigned clusters. The observed approximate cluster silhouette scores were: Cluster 1 ≈ 0.749 , Cluster 2 ≈ 0.000 and Cluster 3 ≈ 0.326 . Cluster 1 therefore showed strong separation, Cluster 3 showed moderate separation, while Cluster 2 had weak separation and may overlap with other customer behaviour.

10. Visualization

A Scatter Plot was used to visualize the customer clusters using Quantity-Sum and UnitPrice-Mean. The visualization helps compare customer groups and observe how purchasing quantity and average unit price differ across clusters.

11. Orange Workflow

The final Orange workflow follows this sequence:

File → Preprocess → Group By → Select Columns → Preprocess → K-Means → Silhouette Plot / Scatter Plot.

A second Group By widget was also used after K-Means to summarize the standardized feature values for each cluster.

12. Conclusion

The project successfully demonstrates customer segmentation using K-Means clustering in Orange Data Mining. Transaction data was transformed into customer-level information, standardized, clustered into three groups and evaluated using silhouette analysis. The resulting segments provide a simple way to distinguish customers according to purchasing volume, activity and average price characteristics.

13. Future Scope

The project can be extended by testing different values of K, adding monetary-value features such as total spending, using RFM (Recency, Frequency and Monetary) analysis, comparing K-Means with hierarchical or DBSCAN clustering, and developing dashboards for business-oriented customer analysis.

14. Project Files

The final submission includes the dataset (Online Retail.xlsx), Orange workflow (Customer_Segmentation_KMeans.ows), presentation (PPTX), and this project report. These files together provide the dataset, implementation, presentation and documentation of the project.